CHANNEL ATTENTION FOR ORIENTED OBJECT DETECTION IN REMOTE SENSING IMAGES

Vaibhav Panchal
Gujarat Technological University, India
vaibhavpanchal1808@gmail.com

Maitrik K. Shah
Gujarat Technological University, India
maitrikshah.ce@gmail.com

***Abstract*—Detecting small objects in remote sensing images is challenging due to complex backgrounds and scale variations. YOLO-based detectors are widely used for efficient end-to-end oriented object detection. In this study, the Efficient Channel Attention (ECA) module is integrated into the YOLOv11s-OBB framework to enhance channel-wise feature representation. The proposed modification is evaluated on the DIOR-R (Rotated-Bounding-Box Aerial Object Detection Dataset) dataset using standard oriented object detection metrics. Experimental results on the DIOR-R dataset demonstrate consistent improvements across all evaluation metrics, including Precision, Recall, (mAP$_{50}$), and (mAP$_{50–95}$), confirming the effectiveness of the proposed ECA-enhanced framework for oriented object detection.**

***Index Terms*—YOLOv11-OBB, remote sensing images, small object detection, channel attention, Efficient Channel Attention.**

I. INTRODUCTION

Object detection in remote sensing imagery plays a critical role in many real-world applications, including urban infrastructure analysis [1], traffic supervision [2], environmental monitoring [3], and security-related [4] tasks. With the rapid advancement of high-resolution aerial and satellite imaging technologies, large-scale remote sensing datasets have become increasingly available, enabling the widespread adoption of deep learning methods for automated object detection. Despite these advances, detecting objects in remote sensing imagery remains challenging due to several intrinsic characteristics of aerial scenes. Compared with natural images, remote sensing data often contain complex backgrounds, large variations in object scale, dense spatial distributions, and objects with arbitrary orientations, all of which significantly increase the difficulty of reliable detection. [5].

To better represent the orientation characteristics of objects in aerial imagery, oriented object detection has emerged as an important research direction. Unlike conventional object detectors that employ horizontal bounding boxes, oriented object detection methods utilize rotated bounding boxes to more accurately capture object orientation and shape. Early studies, such as rotation-based proposal approaches [6–8], demonstrated that explicitly modeling orientation information can significantly improve detection robustness in aerial scenes [6, 8]. Subsequent research introduced feature transformation and alignment strategies to further enhance localization accuracy under complex imaging conditions [7].

Most early oriented object detection methods adopt a two-stage detection paradigm [9, 10, 6]. In the first stage, candidate object regions are generated using region proposal mechanisms, while in the second stage, these proposals are refined through classification and bounding box regression. This two-stage design allows accurate localization and orientation modeling but often incurs higher computational cost and limits efficiency when applied to large-scale remote sensing datasets.

Alongside these developments, one-stage detection paradigms have become increasingly attractive due to their computational efficiency and simplified detection pipelines. Frameworks such as SSD (Single Shot MultiBox Detector) [11] and YOLO [12] directly predict object categories and bounding boxes without a separate proposal stage, making them well suited for large-scale remote sensing datasets [11]. Recent studies adapting YOLO-based architectures for aerial imagery have shown that detection performance is highly influenced by architectural design choices, especially when balancing detection accuracy with computational efficiency. [12].

Despite their efficiency, YOLO-based detectors still face challenges in effectively capturing discriminative features for small and densely distributed objects, which are common in remote sensing imagery. In such scenarios, informative feature channels may not be sufficiently emphasized during feature extraction, which can reduce detection reliability in complex scenes. Attention mechanisms have shown strong potential for improving feature selectivity by enabling neural networks to focus on informative features while suppressing less relevant responses. In particular, channel attention mechanisms adaptively recalibrate feature responses across channels, enhancing the representation of semantically meaningful features with minimal computational overhead. However, despite their effectiveness in many deep learning architectures, the impact of channel attention mechanisms within YOLO-based oriented object detection frameworks remains insufficiently explored. [13–15]

This study addresses these limitations by enhancing the YOLOv11s-OBB architecture through the integration of an Efficient Channel Attention (ECA) [16] module. The ECA mechanism enables adaptive channel-wise feature recalibration, allowing the network to emphasize informative features while suppressing less relevant responses. Due to its lightweight design, the ECA module introduces minimal computational overhead while improving feature representation for small object detection in remote sensing imagery.

To evaluate the effectiveness of the proposed approach, experiments are conducted on the DIOR-R dataset, a large-scale remote sensing benchmark containing oriented bounding box annotations for objects with arbitrary orientations [17]. The baseline YOLOv11s-OBB detector and the ECA-enhanced variant are trained under identical experimental conditions to ensure a fair comparison.

The main contributions of this work are summarized as follows:

- We introduce a lightweight channel attention enhancement for the YOLOv11s-OBB architecture by integrating the Efficient Channel Attention (ECA) module to improve feature representation for oriented object detection.
- We perform a controlled experimental study evaluating the impact of channel attention on small object detection in remote sensing imagery.
- We demonstrate that the proposed ECA-enhanced YOLOv11-OBB improves detection accuracy on the DIOR-R dataset while maintaining computational efficiency.

II. Related Work

Object detection in remote sensing imagery has been widely studied due to its importance in aerial scene understanding. With the development of large-scale remote sensing datasets such as DOTA and DIOR-R, significant progress has been made in detecting objects with arbitrary orientations [5, 17]. Existing research in this area can generally be categorized into oriented object detection frameworks and feature representation enhancement methods.

Early studies focused on oriented object detection techniques that explicitly model object orientation in aerial images. For example, R2CNN introduced rotational region proposals to improve orientation robustness in detection tasks [6]. Subsequent works further explored rotation-aware detection strategies by incorporating rotated bounding boxes into the detection pipeline. Methods based on rotation proposal generation demonstrated the effectiveness of representing objects using oriented bounding boxes to better capture their geometric properties [8]. Later, RoI transformation approaches were proposed to align extracted features with object orientation, enabling improved localization accuracy in complex aerial scenes [7]. These methods established the importance of orientation-aware representations for remote sensing object detection.

Alongside these developments, one-stage object detectors have gained increasing attention due to their efficiency and simplified detection pipelines. Methods such as the Single Shot MultiBox Detector (SSD) directly predict object locations and class labels without requiring region proposal stages, making them suitable for large-scale detection tasks [11]. Building upon this idea, the YOLO family of detectors further advanced one-stage detection by integrating feature extraction and prediction into a unified network architecture. Recent studies adapting YOLO-based detectors for remote sensing imagery have emphasized the importance of architectural design choices in balancing detection accuracy and computational efficiency [12].

Recent studies have investigated various strategies to enhance feature representation in object detection networks. Context-aware detection approaches utilize surrounding scene information to improve detection performance in complex environments. For example, CAD-Net incorporates contextual cues to better detect objects in remote sensing imagery [10]. Other works have explored feature refinement techniques to strengthen discriminative feature representations and improve robustness in challenging scenes [18]. Additionally, advanced feature aggregation and upsampling methods have been proposed to preserve spatial details during multi-scale feature fusion. For instance, CARAFE introduces a content-aware feature reassembly mechanism that improves feature reconstruction during upsampling operations [19].

Attention mechanisms have also been adopted to improve feature representation in deep neural networks. Channel attention modules such as Squeeze-and-Excitation networks and Efficient Channel Attention networks enable adaptive reweighting of feature channels, improving the network's ability to focus on informative features while suppressing irrelevant responses.

Despite these advances, the integration of lightweight channel attention mechanisms within YOLO-based oriented object detection frameworks has received limited investigation. In particular, the influence of channel-wise attention on small object detection in remote sensing imagery remains insufficiently explored. To address this gap, this work investigates the integration of an ECA module into the YOLOv11s-OBB architecture and evaluates its impact on oriented object detection performance.

III. Methodology

This section presents the experimental procedure adopted to examine the role of channel-level feature modulation in a YOLO-based oriented object detection model. A controlled evaluation strategy is employed, by augmenting the baseline architecture with the Efficient Channel Attention (ECA) mechanism and its influence on detection outcomes is assessed.

A. Baseline Detector

The baseline model employed in this study is YOLOv11s-OBB [12], a one-stage oriented object detector designed to predict rotated bounding boxes directly from multi-scale feature representations. The architecture consists of three main components: a convolutional backbone for feature extraction, a feature aggregation neck for multi-scale representation, and an oriented detection head responsible for classification and bounding box regression. This baseline configuration serves as the reference model for all experiments.

B. Efficient Channel Attention Integration

To enhance the quality of feature representations, a channel-level modulation strategy is adopted within the baseline detection network. This strategy operates by modeling relationships among feature channels using a compact convolutional operation, allowing channel importance to be learned without introducing dimensional compression or significant computational overhead. Through this process, channels carrying informative semantic cues are strengthened, whereas less contributive responses are comparatively suppressed. This mechanism is particularly beneficial for remote sensing imagery, where small objects often exhibit weak visual signatures and may be easily obscured by complex background patterns.

In the proposed configuration, this channel modulation mechanism is embedded into selected convolutional layers of the YOLOv11s-OBB network to adjust intermediate feature responses. All remaining components of the architecture are preserved in their original form, ensuring that any observed performance variations can be attributed solely to the inclusion of this channel-level modulation approach.

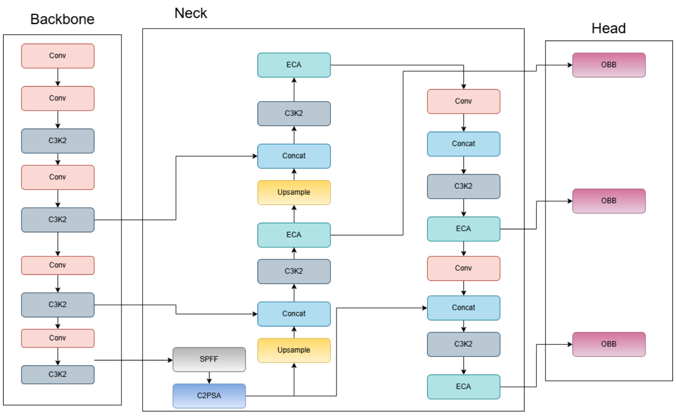

Fig. 1. YOLO11-ECA architecture

Figure 1 presents the modified YOLOv11s-OBB architecture with the integrated ECA module. The attention module is embedded within selected convolutional layers of the backbone to refine intermediate feature representations without altering the overall network structure.

C. Efficient Channel Attention Module

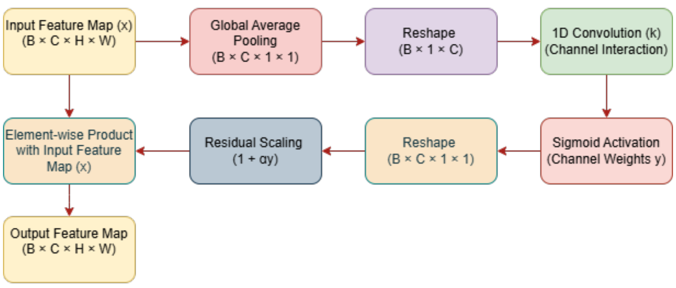

Fig. 2. Block diagram of the Efficient Channel Attention (ECA) module.

Figure 2 illustrates the internal structure of the Efficient Channel Attention (ECA) module. It captures local cross-channel interactions using global average pooling followed by a lightweight one-dimensional convolution and sigmoid activation, enabling adaptive channel-wise feature recalibration.

Let $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$ denote the input feature map, where B, C, H, and W represent the batch size, number of channels, height, and width, respectively. Figure 2 illustrates the workflow of the Efficient Channel Attention (ECA) module, which enhances feature representation by modeling local cross-channel dependencies in a lightweight manner.

As shown in the figure, the process begins with global Average Pooling, which aggregates spatial information across each channel to produce a compact channel descriptor:

$$z_c = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} X_c(i,j), \quad c = 1,\ldots,C. \tag{1}$$

The resulting channel descriptor $\mathbf{z} = [z_1, z_2, \ldots, z_C]$ is then fed into a one-dimensional convolution layer that captures local channel interactions without dimensionality reduction:

$$\mathbf{s} = \sigma\left(\mathrm{Conv1D}(\mathbf{z}, k)\right), \tag{2}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function and k is the kernel size of the convolution.

The kernel size k is adaptively determined based on the channel dimension to balance receptive field size and computational efficiency:

$$k = \left|\frac{\log_2(C) + b}{\gamma}\right|_{\mathrm{odd}}, \tag{3}$$

where γ and b are predefined constants and $|\cdot|_{\mathrm{odd}}$ ensures that k is an odd integer.

Finally, the attention weights are reshaped and applied to the input feature map using a residual-safe scaling operation:

$$\tilde{X}_c = X_c \cdot (1 + \alpha \cdot s_c), \tag{4}$$

where α controls the strength of the attention mechanism and $\tilde{X}_c$ represents the refined output feature of the c-th channel.

IV. Experimental Setup

A. Dataset Description

The DIOR-R [17] dataset contains over 18,000 high-resolution aerial images annotated with oriented bounding boxes across diverse object categories. It includes small and densely distributed objects such as vehicles, ships, and storage tanks, appearing under complex backgrounds and varying scales in urban, rural, and natural scenes.

The dataset exhibits significant variations in object scale, orientation, and spatial distribution, making it suitable for evaluating oriented object detection performance. Following the official split, 18,770 images are used for training and 2,347 for testing. All images are resized to 640 × 640 while preserving the original oriented annotations.

B. Implementation Details

The experiments are conducted using PyTorch with the Ultralytics YOLO training framework. The YOLOv11s-OBB detector is trained for 100 epochs with a batch size of 4 using the AdamW optimizer and an initial learning rate of 1×10^{-3}. All images are resized to 640 × 640, and standard data augmentation is applied during training.

To ensure a fair comparison, the ECA module is integrated only into selected backbone layers, while the remaining architecture is unchanged. Training is performed using AMP on a single GPU.

TABLE I
TRAINING CONFIGURATION

Parameter	Value
Epochs	100
Batch size	4
Optimizer	AdamW
Initial learning rate	0.001
Image size	640
AMP	Enabled
GPU	6 GB NVIDIA 4050

V. RESULTS AND DISCUSSION

This subsection reports a quantitative comparison between the baseline YOLOv11s-OBB detector and its ECA-enhanced counterpart on the DIOR-R dataset.

Detection performance is assessed using Precision (P), Recall (R), mAP_{50}, and $mAP_{50\text{–}95}$, as defined in previous Section. The numerical results obtained for both configurations are summarized in Table II.

As observed from Table II, the introduction of the Efficient Channel Attention module leads to uniform improvements across all reported metrics. Precision shows an increase of 2.6 points, suggesting more reliable classification behavior, while Recall improves by 4.3 points, indicating enhanced detection coverage. Gains are also evident in localization performance, with mAP_{50} and $mAP_{50\text{–}95}$ increasing by 3.8 and 3.4 points, respectively.

In addition to the baseline comparison, Table II includes recent approaches such as RC-YOLO and YOLOv11s-AMMFN. RC-YOLO enhances detection performance using C3k2-RE blocks and CARAFE-based feature upsampling, focusing on improving feature representation for small objects. YOLOv11s-AMMFN introduces advanced modules such as AMCC and DHEM to strengthen feature fusion and contextual modeling. Compared to these methods, the proposed YOLOv11s-OBB + ECA achieves competitive or superior performance while maintaining a significantly lower computational overhead and preserving the simplicity of the original architecture.

Fig. 3 presents qualitative comparisons between Ground Truth, baseline YOLOv11s-OBB [12], and the proposed YOLOv11s-OBB + ECA on DIOR-R [17]. In the first row

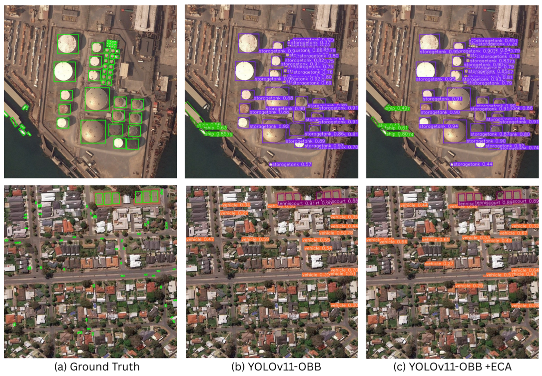

Fig. 3. Detection results on the DIOR-R dataset. (a) Ground Truth, (b) baseline YOLOv11s-OBB, and (c) proposed YOLOv11s-OBB + ECA.

(dense storage tanks), the baseline produces overlapping, redundant, and low-confidence detections, whereas the ECA-enhanced model generates more accurate and confident bounding boxes with minimal duplication. In the second row (urban residential areas), the baseline misses small objects and shows inconsistent predictions, while the proposed model achieves better localization and more consistent detection of small and closely spaced objects. Overall, the integration of ECA improves feature discrimination, enabling robust performance in dense scenes, complex backgrounds, and varying object scales.

Fig. 4. Zoomed-in comparison of detection results. The proposed YOLOv11s-OBB + ECA demonstrates improved localization accuracy and reduced false detections in dense regions compared to the baseline.

To further highlight the effectiveness of the proposed method, Fig. 4 presents zoomed-in views of challenging regions. It can be observed that the baseline model suffers from overlapping predictions, missed detections, and lower confidence scores in densely populated areas. In contrast, the proposed YOLOv11s-OBB + ECA produces more accurate and consistent detections with clearer object boundaries. The improvements are particularly evident for small and closely spaced objects, where the proposed method significantly reduces ambiguity and enhances detection reliability.

Notably, the ECA module functions without reducing dimensionality and adds only a minimal number of extra pa-

TABLE II
EVALUATION RESULTS ON THE DIOR-R DATASET

Model	**Modification**	**Dataset**	**P**	**R**	**mAP$_{50}$**	**mAP$_{50–95}$**	**Time (ms)**	**Params (M)**	**GFLOPs**
YOLOv11s-OBB (Baseline)	–	DIOR-R	82.6	75.4	81.5	62.5	20.6	9.72	22.6
RC-YOLO [20]	C3k2-RE + CARAFE	DIOR-R	**89.16**	75.27	81.67	–	–	–	–
YOLOv11s-AMMFN [21]	AMCC + DHEM	DIOR-R	81.1	75.4	80.5	59.8	30.7	18.45	36.6
YOLOv11s-OBB + ECA	ECA	DIOR-R	85.2	**79.7**	**85.3**	**65.9**	21.7	9.72	22.9

rameters, thus maintaining the original detection architecture and inference characteristics. Consequently, the performance improvements observed are due to better feature selectivity rather than increased architectural complexity.

VI. CONCLUSION

In this research, we investigate the impact of incorporating a lightweight channel attention mechanism into the YOLOv11s-OBB[12] framework for detecting oriented objects in remote sensing images. By integrating the Efficient Channel Attention (ECA) module into the baseline detector, our proposed method improves channel-wise feature representation while maintaining the original network architecture. Experimental results on the DIOR-R [17] dataset show consistent enhancements in various evaluation metrics, such as Precision, Recall, and mean Average Precision, indicating better detection completeness and localization accuracy. These results imply that channel-wise feature recalibration allows the model to more effectively utilize discriminative information, which is especially advantageous for identifying small and arbitrarily oriented objects in complex remote sensing environments. Notably, the proposed modification adds minimal architectural complexity and retains the efficiency of the baseline detector, making it suitable for practical remote sensing applications. Future research will explore other attention mechanisms, different integration strategies, and broader evaluations on larger-scale remote sensing datasets.

REFERENCES

[1] V. Mnih and G. E. Hinton, "Learning to detect roads in high-resolution aerial images," in *Proceedings of the European Conference on Computer Vision (ECCV).* Springer, 2010, pp. 210–223.

[2] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.

[3] Z. Zhu and C. E. Woodcock, "Continuous monitoring of land disturbance using landsat time series," *Remote Sensing of Environment*, vol. 144, pp. 152–171, 2014.

[4] W. Li, V. Mahadevan, and N. Vasconcelos, "Anomaly detection and localization in crowded scenes," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 1, pp. 18–32, 2013.

[5] G.-S. Xia, X. Bai, J. Ding, Z. Zhu, S. Belongie, J. Luo, M. Datcu, M. Pelillo, and L. Zhang, "Dota: A large-scale dataset for object detection in aerial images," *arXiv preprint arXiv:1711.10398*, 2019. [Online]. Available: http://arxiv.org/abs/1711.10398

[6] Y. Jiang, X. Zhu, X. Wang, S. Yang, W. Li, H. Wang, P. Fu, and Z. Luo, "R2cnn: Rotational region cnn for orientation robust scene text detection," *arXiv preprint arXiv:1706.09579*, 2017. [Online]. Available: http://arxiv.org/abs/1706.09579

[7] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning roi transformer for detecting oriented objects in aerial images," *arXiv preprint arXiv:1812.00155*, 2018. [Online]. Available: http://arxiv.org/abs/1812.00155

[8] J. Ma, W. Shao, H. Ye, L. Wang, H. Wang, Y. Zheng, and X. Xue, "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Transactions on Multimedia*, 2018. [Online]. Available: http://arxiv.org/abs/1703.01086

[9] R. Girshick, "Fast r-cnn," 9 2015. [Online]. Available: http://arxiv.org/abs/1504.08083

[10] G. Zhang, S. Lu, and W. Zhang, "Cad-net: A context-aware detection network for objects in remote sensing imagery," *arXiv preprint arXiv:1903.00857*, 2019. [Online]. Available: http://arxiv.org/abs/1903.00857

[11] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "Ssd: Single shot multibox detector," 12 2016. [Online]. Available: http://arxiv.org/abs/1512.02325http://dx.doi.org/10.1007/978-3-319-46448-0_2

[12] R. Khanam and M. Hussain, "Yolov11: An overview of the key architectural enhancements," *arXiv preprint arXiv:2410.17725*, 2024. [Online]. Available: http://arxiv.org/abs/2410.17725

[13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.

[14] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7132–7141.

[15] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "Cbam: Convolutional block attention module," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 3–19.

[16] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "Eca-net: Efficient channel attention for deep convolutional neural networks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 11 534–11 542.

[17] K. Li, G. Wan, and G. Cheng, "Dior," 2025. [Online]. Available: https://dx.doi.org/10.21227/tq1e-nx82

[18] D. Shi, C. Zhao, J. Shao, M. Feng, L. Luo, B. Ouyang, and J. Huang, "Context-aware enhanced feature refinement for small object detection with deformable detr," *Frontiers in Neurorobotics*, vol. 19, 2025. [Online]. Available: https://www.frontiersin.org/articles/10.3389/fnbot.2025.1588565/full

[19] J. Wang, K. Chen, R. Xu, Z. Liu, C. C. Loy, and D. Lin, "Carafe: Content-aware reassembly of features," *arXiv preprint arXiv:1905.02188*, 2019. [Online]. Available: http://arxiv.org/abs/1905.02188

[20] H. Zhou, W. Yang, X. Yang, and J. Ren, "Rc-yolo: A modified yolo network for small oriented target detection in rs images," Tech. Rep., 2025.

[21] J. Qu, Z. Tang, L. Zhang, Y. Zhang, and Z. Zhang.